

White Paper 3 — Scoring, ranking, and lead selection

Pipeline step: Score and rank druggable targets; select leads **Inputs:** `druggability_annotation.csv`, `target_universe.csv`
Outputs: `druggable_target_ranking.csv`, `fig_target_ranking.png`, `lead_rationale.md`

1. Purpose

This step fuses the disease-genetics evidence (White Paper 1) and the druggability evidence (White Paper 2) into a single ranking, then applies a mechanistic filter to select the three leads carried into structure-based work: **STAT3, RORC (RORyt), and JAK3**.

2. The composite druggability score

Each of the 154 annotated targets received a composite score combining seven normalized terms across three evidence blocks:

```
druggability_score =  
  DE:          0.18 s_effect + 0.12 s_signif  
  pathway:     0.15 s_pathway  
  druggability: 0.16 s_tractability + 0.12 s_chem + 0.12 s_drugs  
  structure:   0.15 s_struct
```

Each `s_*` is a min-max-normalized feature: `s_effect` / `s_signif` from the meta-analysis, `s_pathway` from leading-edge/regulon membership, `s_tractability` from the Open Targets bucket [1], `s_chem` from ChEMBL potent-inhibitor count [2], `s_drugs` from known-drug count, and `s_struct` from availability of a ligand-bound experimental structure. The weighting deliberately balances "is this gene important in psoriasis" against "can this protein be drugged", so that neither a strong-but-undruggable nor a druggable-but-irrelevant target dominates.

3. Why a raw score is not enough

The raw top of the ranking included targets whose tractability derives from drug programs unrelated to psoriasis (histamine and serotonin receptors, renin). A high score built on off-axis chemistry is a false lead. We therefore filtered the ranked pool by three simultaneous conditions:

1. **On a psoriasis disease axis** (mechanistic relevance);
2. **Structure-ready** (a ligand-bound experimental structure exists);
3. **Tractability bucket $\geq$ Phase-1/structure level** (real small-molecule precedent).

The concept of restricting to privileged, mechanistically-grounded targets follows directly from the druggable-genome framework [3].

4. The three leads and their rationale

Applying the filter and reading the mechanism gives a coherent **JAK-STAT / Th17 signaling axis**:

Lead	Role in psoriasis	Direction	Tractability	Potent inhibitors
JAK3	Upstream kinase feeding cytokine-receptor signaling	up (+1.55)	Approved-drug class	6,069
STAT3	Transcription factor linking keratinocytes and immune infiltrate	up (+1.25)	Discovery precedence	112
RORC / RORyt	Master transcription factor of the Th17 lineage	down (-2.47)	Phase-1 small molecule	9,766

The axis is mechanistically continuous: **JAK3** → **STAT3** → **RORyt** → **IL-17**. JAK3 was designated the most classically tractable node (approved-drug kinase class); RORyt the best tractable transcription factor with a genuine ligand-binding pocket; STAT3 the user's designated target of primary interest. STAT3 ranked 43rd on the raw composite but was retained by design — it is the biological anchor of the project — and its inclusion is transparently a mechanistic rather than a score-driven choice.

5. Deliverables

- `druggable_target_ranking.csv` — all 154 targets with per-block sub-scores and the composite.
- `fig_target_ranking.png` — ranked bar chart with the three leads highlighted.
- `lead_rationale.md` — narrative justification for the three-lead selection.

6. Caveats

- STAT3's retention is a deliberate override of the pure score; this is stated openly rather than hidden by re-weighting.
- The score is a prioritisation aid, not a ground truth; secondary candidates (NOS2, MMP9/12, IDO1, PLK1/AURKA) remain in the table for follow-up.

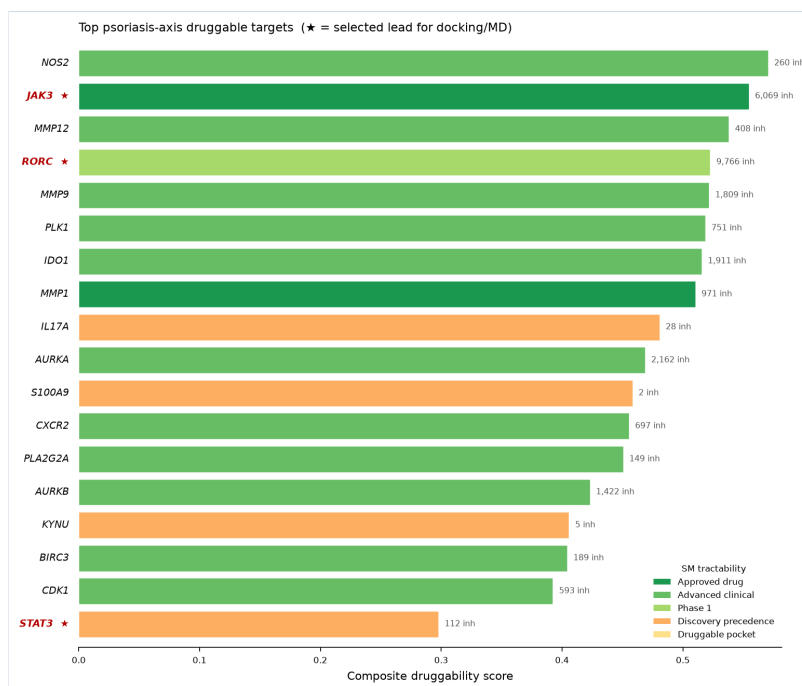


Figure: key output of this pipeline step.

References

1. Ochoa et al. The next-generation Open Targets Platform: reimagined, redesigned, rebuilt. *Nucleic Acids Research*. 2022. doi:10.1093/nar/gkac1046
2. Zdrazil et al. The ChEMBL Database in 2023: a drug discovery platform spanning multiple bioactivity data types and time periods. *Nucleic Acids Research*. 2023. doi:10.1093/nar/gkad1004
3. Hopkins et al. The druggable genome. *Nature Reviews Drug Discovery*. 2002. doi:10.1038/nrd892